

TW6 AaroHi Kondpalle

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PS C:\Users\aaroh> kubectl get nodes
NAME          STATUS    ROLES    AGE   VERSION
minikube      Ready    control-plane  25d   v1.35.1
PS C:\Users\aaroh> minikube addons enable metrics-server
* metrics-server is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNER
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- Using image registry.k8s.io/metrics-server/metrics-server:v0.8.1
* The 'metrics-server' addon is enabled
PS C:\Users\aaroh> kubectl get pods -n kube-system
NAME                                READY   STATUS    RESTARTS   AGE
coredns-7d764666f9-8sl2c           1/1     Running   5 (23d ago)  25d
etcd-minikube                      1/1     Running   5 (23d ago)  25d
kube-apiserver-minikube             1/1     Running   5 (23d ago)  25d
kube-controller-manager-minikube    1/1     Running   6 (23d ago)  25d
kube-proxy-snp5h                   1/1     Running   5 (23d ago)  25d
kube-scheduler-minikube            1/1     Running   5 (23d ago)  25d
metrics-server-9d74bb658-jvr5s     1/1     Running   0           70s
storage-provisioner                 1/1     Running   11 (73s ago)  25d
PS C:\Users\aaroh> kubectl top nodes
NAME          CPU(cores)   CPU(%)   MEMORY(bytes)   MEMORY(%)
minikube      633m        5%       1982Mi          25%
PS C:\Users\aaroh> mkdir C:\Users\aaroh\k8s-autoscaling

Directory: C:\Users\aaroh
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PS C:\Users\aaroh> cd C:\Users\aaroh\k8s-autoscaling
PS C:\Users\aaroh\k8s-autoscaling> notepad deployment.yaml
PS C:\Users\aaroh\k8s-autoscaling> kubectl apply -f deployment.yaml
deployment.apps/social-media-app created
PS C:\Users\aaroh\k8s-autoscaling> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
alertmanager-monitoring-kube-prometheus-alertmanager-0  2/2     Running   8 (23d ago)  25d
loki-0                                1/1     Running   5 (23d ago)  25d
loki-promtail-7c7t4                  1/1     Running   4 (23d ago)  25d
monitoring-grafana-547fcbddc9-hpr22  2/3     CrashLoopBackOff  47 (50s ago)  25d
monitoring-kube-prometheus-operator-8876fbff5-6jnp2    1/1     Running   6 (2m43s ago)  25d
monitoring-kube-state-metrics-6cbbf66fff-qj6tn         1/1     Running   7 (2m41s ago)  25d
monitoring-prometheus-node-exporter-hbrbk              1/1     Running   4 (23d ago)  25d
nginx-56c45fd5ff-87q74                 1/1     Running   5 (23d ago)  25d
prometheus-monitoring-kube-prometheus-prometheus-0    2/2     Running   8 (23d ago)  25d
social-media-app-c859d545-p7xwc         1/1     Running   0           14s
PS C:\Users\aaroh\k8s-autoscaling> notepad service.yaml
PS C:\Users\aaroh\k8s-autoscaling> kubectl apply -f service.yaml
service/social-media-service created
PS C:\Users\aaroh\k8s-autoscaling> kubectl get service
NAME                                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)
alertmanager-operated              ClusterIP    None           <none>        9093/TCP,9094/TCP,909
4/UDP 25d
kubernetes                         ClusterIP    10.96.0.1     <none>        443/TCP
loki                               ClusterIP    10.110.210.164 <none>        3100/TCP
loki-headless                     ClusterIP    None          <none>        3100/TCP
loki-memberlist                   ClusterIP    None          <none>        7946/TCP
monitoring-grafana                 ClusterIP    10.97.138.71  <none>        80/TCP
monitoring-kube-prometheus-alertmanager  ClusterIP    10.108.179.96 <none>        9093/TCP,8080/TCP
monitoring-kube-prometheus-operator  ClusterIP    10.106.187.61 <none>        443/TCP
monitoring-kube-prometheus-prometheus  ClusterIP    10.101.178.195 <none>        9090/TCP,8080/TCP
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prometheus-operated          ClusterIP   None          <none>          9090/TCP
25d
social-media-service         NodePort    10.98.118.26  <none>          80:30577/TCP
6s
PS C:\Users\aaroh\k8s-autoscaling> minikube service social-media-service

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NAMESPACE	NAME	TARGET PORT	URL
default	social-media-service	80	http://192.168.49.2:30577

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* Starting tunnel for service social-media-service.

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NAMESPACE	NAME	TARGET PORT	URL
default	social-media-service		http://127.0.0.1:53184

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* Opening service default/social-media-service in default browser...
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
* Stopping tunnel for service social-media-service.
PS C:\Users\aaroh\k8s-autoscaling> kubectl get service social-media-service
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
social-media-service NodePort    10.98.118.26 <none>        80:30577/TCP     88s
PS C:\Users\aaroh\k8s-autoscaling> kubectl get endpoints social-media-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME                ENDPOINTS          AGE
social-media-service 10.244.0.53:80     104s
PS C:\Users\aaroh\k8s-autoscaling> kubectl port-forward service/social-media-service 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80

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deployment.yaml

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service.yaml

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apiVersion: apps/v1
kind: Deployment
metadata:
  name: social-media-app
spec:
  replicas: 1
  selector:
    matchLabels:
      app: social-media-app
  template:
    metadata:
      labels:
        app: social-media-app
    spec:
      containers:
        - name: social-media-app
          image: nginx:latest
          ports:
            - containerPort: 80
      resources:
        requests:
          cpu: "100m"
        limits:
          cpu: "500m"

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```
deployment.yaml service.yaml
File Edit View

apiVersion: v1
kind: Service
metadata:
  name: social-media-service
spec:
  selector:
    app: social-media-app
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
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edia-service; done"
pod/load-generator created
PS C:\Users\aaroh\k8s-autoscaling> kubectl get hpa -w
NAME          REFERENCE          TARGETS          MINPODS  MAXPODS  REPLICAS  AGE
social-media-hpa  Deployment/social-media-app  cpu: 0%/50%      1         5         1         80s
social-media-hpa  Deployment/social-media-app  cpu: 54%/50%     1         5         1        101s
```

```
social-media-app-c859d545-p7xwc 1/1 Running 0 9m4s
PS C:\Users\aaroh> kubectl get pods -l app=social-media-app -w
NAME          READY  STATUS    RESTARTS  AGE
social-media-app-c859d545-hnxtj 1/1    Running   0          6s
social-media-app-c859d545-p7xwc 1/1    Running   0         9m37s
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edia-service; done"
pod/load-generator created
PS C:\Users\aaroh\k8s-autoscaling> kubectl get hpa -w
NAME          REFERENCE          TARGETS          MINPODS  MAXPODS  REPLICAS  AGE
social-media-hpa  Deployment/social-media-app  cpu: 0%/50%      1         5         1         80s
social-media-hpa  Deployment/social-media-app  cpu: 54%/50%     1         5         1        101s
social-media-hpa  Deployment/social-media-app  cpu: 75%/50%     1         5         1       2m39s
social-media-hpa  Deployment/social-media-app  cpu: 75%/50%     1         5         2       2m53s
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